AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-07-31 | Entry window 31 July 2026 | 11 records in scope

11 RECORDS IN SCOPE (7 SUBTOPICS)	**8** EFFECT ESTIMATES EXTRACTED	**4** RECORDS FROM CHINA (36% OF CORPUS)	**2** EXPOSURE ASSESS-MENT & MODELLING	**0** SENSING / INSTRUMENTATION

Signal of the day

Road-network proxies fail against personal $PM_{2.5}$ in sub-Saharan Africa — and they fail with the wrong sign, not merely weakly.

Nyoni et al. (*Air Qual Atmos Health*) validated three of the exposure proxies that epidemiology routinely substitutes for measurement — population-weighted road network density, Euclidean distance to highway, Euclidean distance to main road — against **a year of filter-based personal $PM_{2.5}$ from 343 postpartum participants** in The Gambia, Kenya and Mozambique. Weighted road density correlated with personal $PM_{2.5}$ only in Mozambique ($\rho = 0.351$, $p = 0.005$); Kenya and The Gambia were null ($\rho = -0.041$ and -0.020). Distance-to-highway was *positive* in The Gambia and Mozambique but **significantly negative in Kenya** ($\rho = -0.224$, $p = 0.018$) — the proxy inverts across countries. Single-variable random forests never exceeded $R^2 = 0.45$; combining all three reached $R^2 = 0.52$ (RMSE 31.7 $\mu g/m^3$) in Kenya and 0.60 (8.9 $\mu g/m^3$) in Mozambique, with a global model at 0.46. An RMSE of 30 $\mu g/m^3$ is larger than the entire annual mean in most European cohorts. The operational reading for distributed sensing is direct: in settings where household energy and biomass burning dominate the personal exposure budget, traffic geometry is not a transferable surrogate, and the case for physically deploying low-cost nodes rather than regressing on road maps is strongest exactly where reference infrastructure is thinnest.

What changed today

- **Two independent records say the exposure variable is the problem, not the model.** Nyoni et al. show road-geometry proxies inverting in sign against measured personal $PM_{2.5}$ between African countries; Knapova et al. show that step counts fall 361 per 1-SD PM_{10} rise, which makes time-activity patterns *endogenous* to concentration. Taken together they bracket the same failure from both ends: the spatial surrogate is wrong, and the mobility assumption that would let you correct it is wrong too.
- **PM mass was the null term in its own model, twice.** Zhu et al. report $PM_{2.5}$ RR 1.004 (0.991–1.017) and PM_{10} 1.004 (0.999–1.010) per 10 $\mu g/m^3$ for pneumonia admissions against NO_2 at 1.221 (1.174–1.271); Yue et al. report no particulate term at all and attribute childhood bronchopneumonia to NO_2 and SO_2. In both cases the pollutants share a combustion source and a seasonal cycle and no multi-pollutant model is fitted, so these are statements about correlation structure rather than about which pollutant acts — but the operational consequence stands: a mass-only monitor would have recorded nothing on either study's strongest days.
- **Mixture methods are outrunning the sample sizes they are applied to.** Chen et al. run quantile g-computation over 29 hazards in 7,092 adults and recover asthma odds of 1.35 (1.06–1.73) per quintile **only** in high-social-vulnerability tracts; Sezer et al. run a six-pollutant WQS index on 127 cleft cases and report ozone dominance with no bootstrap weight distribution; Zhou et al. run WQS on urinary metabolites measured *at or after* diagnosis. The technique is the same; the credibility is not, and it tracks sample size and exposure timing rather than the model.
- **Zero sensing or instrumentation records again.** This is a harvest defect, not a real absence: the July monthly rollup recovered 23 sensing records for the month against 11 across all five daily issues combined. The pipeline still has no date-windowed channel into *Atmos. Meas. Tech.* or *Atmos. Chem. Phys.* See the provenance box.

Corpus analytics

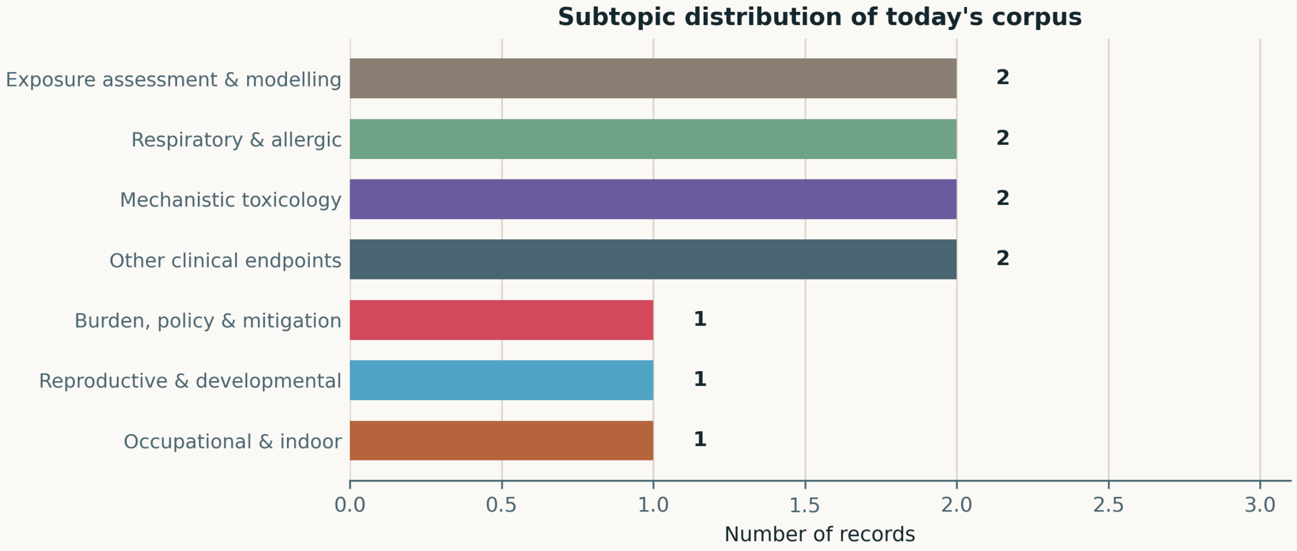

Fig. Subtopic assignment of the 11 in-scope records, single-label by dominant contribution. Seven of the ten subtopics are populated; Neuro / mental health, Cardiovascular & metabolic and Sensing, forecasting & instrumentation are empty this window. Colours are fixed across issues so a subtopic keeps its hue in the rollups.

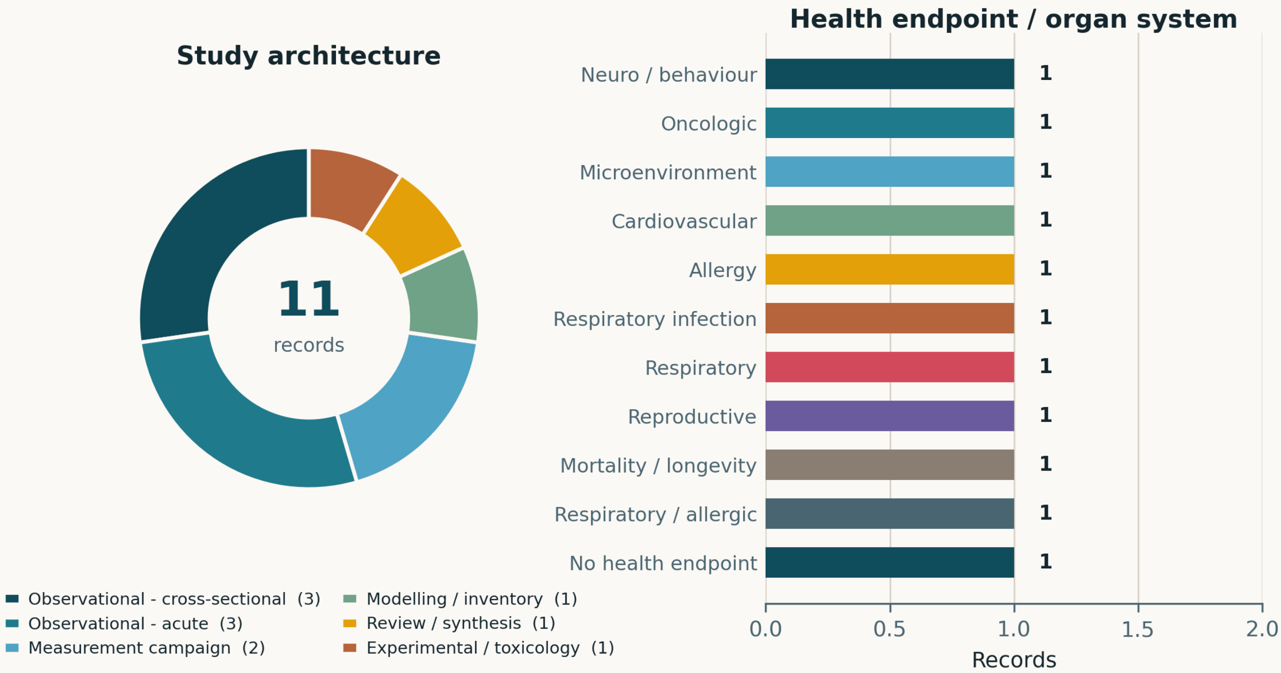

Fig. Left — study architecture. Observational designs account for 6/11, split 3 cross-sectional (two case-control, one exposome-wide) and 3 acute time-series or panel. Two measurement campaigns, one modelling record, one evidence synthesis and one experimental-toxicology record complete the set. Right — health endpoint. Two records carry no human health outcome: the exposure-proxy validation (plotted as *No health endpoint*) and the indoor-air characterisation (plotted as *Microenvironment*).

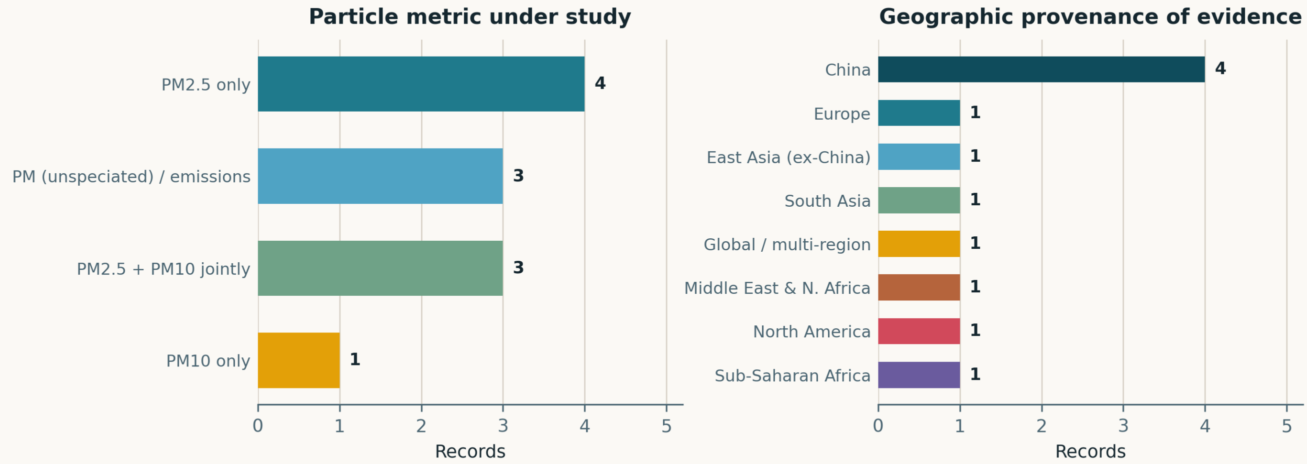

Fig. Left — particle metric. $PM_{2.5}$ alone in 4 records, $PM_{2.5}$ with PM_{10} in 3, PM_{10} alone in 1, and 3 records use an unspeciated or non-ambient particulate metric (diesel SRM 1650b, indoor construction PM, traffic gases standing in for PM). Right — geography. China supplies 4 of 11 records; the single sub-Saharan African record is the exposure-proxy validation study.

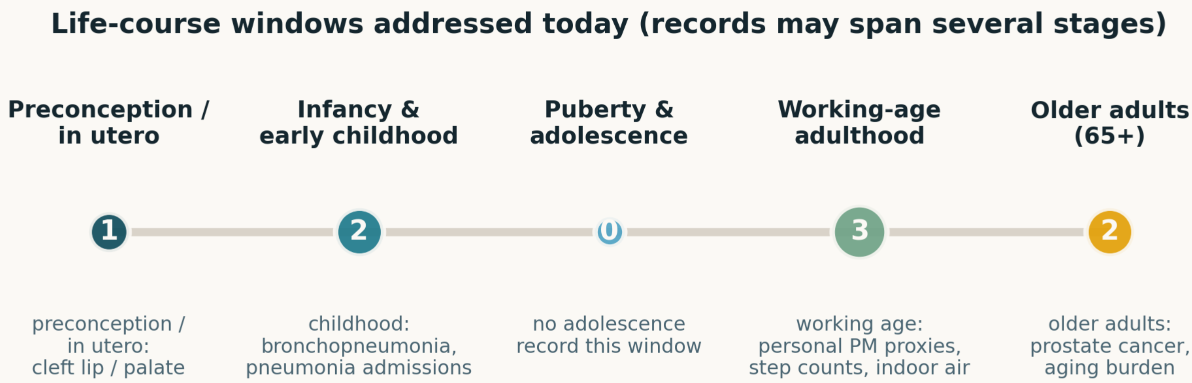

Fig. Life-course windows addressed. Counts are non-exclusive. Working age leads at 3; no record addresses adolescence. Records with no identifiable human life-course window — the ex-vivo cardiac preparation and the rodent meta-analysis — are excluded from the tally.

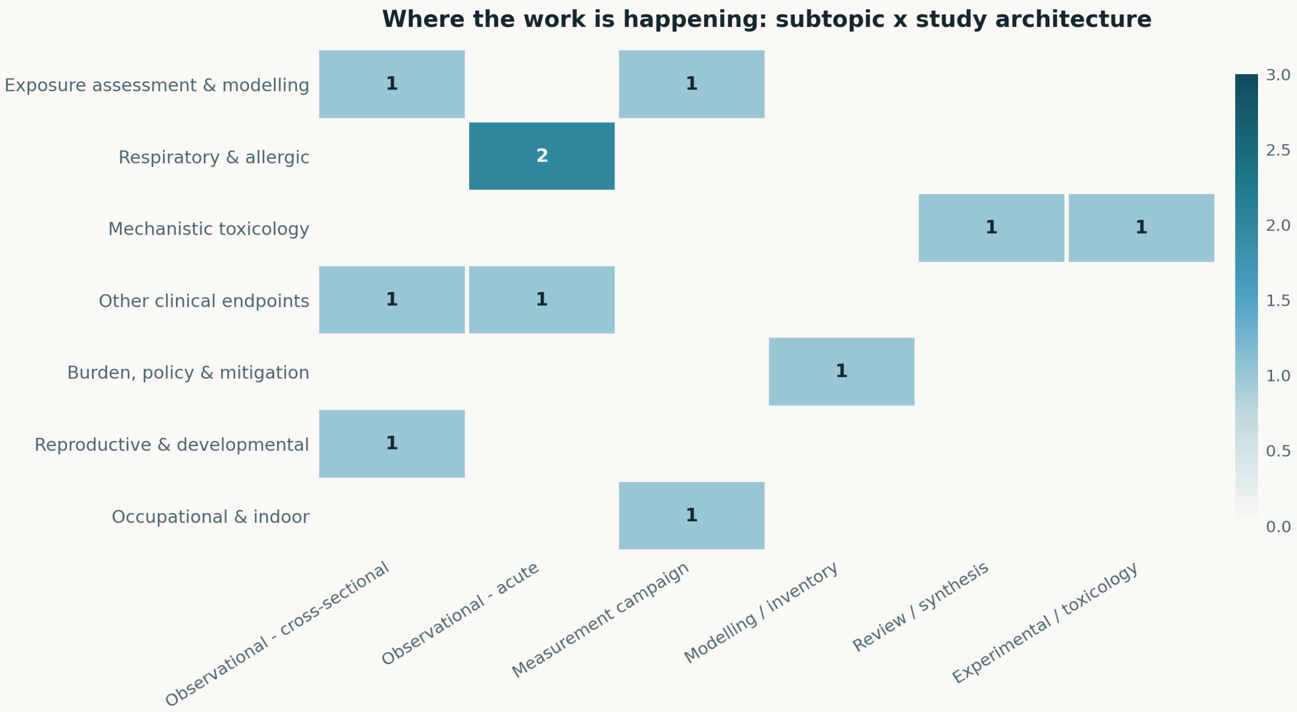

Fig. Subtopic × study-architecture cross-tabulation. With 11 records the matrix is sparse and little should be read from individual cells; the one structural feature worth noting is the empty Sensing row, which is a harvest artefact rather than a property of the literature.

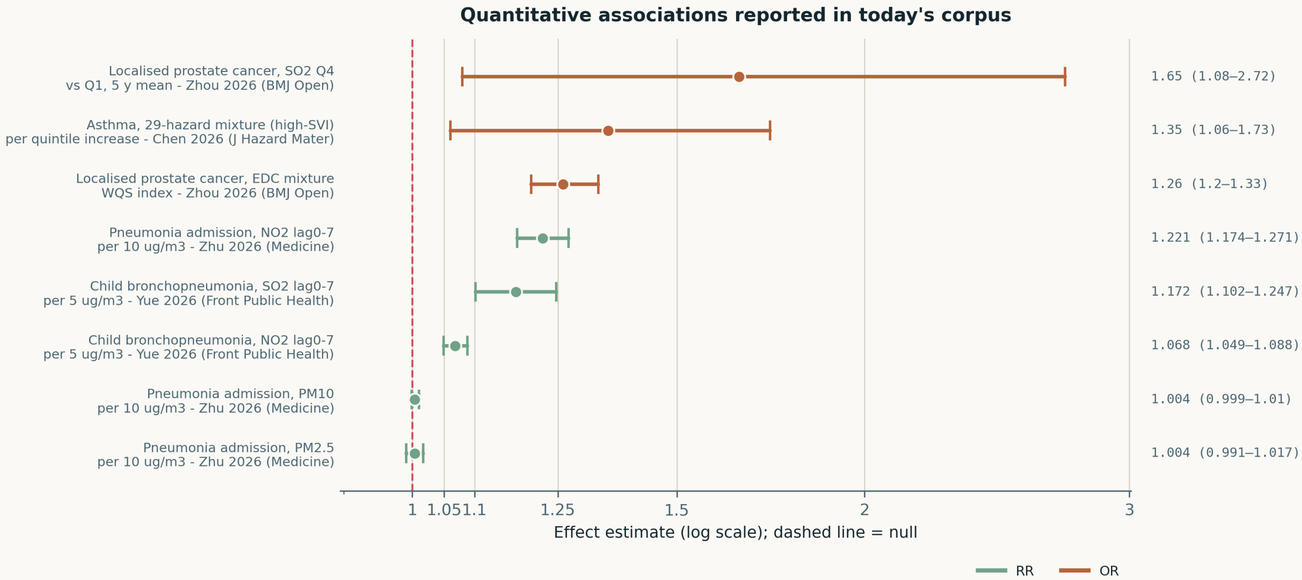

Fig. The eight quantitative effect estimates in the corpus on a common ratio axis. **The exposure contrasts are not commensurable** and the plot must not be read as a meta-analysis: increments range from per 5 $\mu g/m^3$ to per 10 $\mu g/m^3$ to quartile and quintile contrasts to a WQS index, and the outcome scale mixes RR and OR. Note that the two PM-specific estimates (Zhu et al., $PM_{2.5}$ and PM_{10} pneumonia admissions) are the smallest in the set and are null or borderline, while every large estimate belongs to a gaseous co-pollutant or a mixture index.

Paper-level digest

Exposure assessment & modelling 2 records

Nyoni et al. 2026 Air Qual Atmos Health

Performance of road-traffic-based exposure proxies against personal $PM_{2.5}$ measurements in three sub-Saharan African countries

One year of filter-based outdoor personal $PM_{2.5}$ from 343 postpartum participants in The Gambia, Kenya and Mozambique, used to validate population-weighted road network density (wRND), Euclidean distance to highway (EH) and to main road (EM). Spearman correlations were weak and **sign-inconsistent across countries**: wRND significant only in Mozambique ($\rho = 0.351$), EH positive in The Gambia (0.335) and Mozambique (0.292) but **negative in Kenya** (-0.224, $p = 0.018$). Single-proxy random forests gave $R^2 < 0.45$ everywhere; combined models reached $R^2 = 0.52$ (RMSE 31.7 $\mu g/m^3$) in Kenya and 0.60 (8.9 $\mu g/m^3$) in Mozambique. The sampled population — postpartum women — is not representative of general mobility patterns, and only outdoor-filtered personal exposure was measured, so the household cooking contribution that most plausibly drives the proxy failure is inferred rather than partitioned. Still the strongest available empirical case that road-geometry proxies require local validation before use, and a direct argument for node deployment over GIS surrogacy.

doi: 10.1007/s11869-026-02060-y

Chen et al. 2026 J Hazard Mater

Geospatial-based neighborhood hazards and asthma in adults: interaction with social vulnerability

Exposome-wide association study over 29 residential hazards — air pollutants, industrial VOC emissions, highway density and proximity to 13 pollution-source classes — in 7,092 North Carolina adults (2013–2022), stratified at the median census-tract Social Vulnerability Index, with quantile g-computation for the mixture. In the overall population only proximity to dry-cleaning-solvent-contaminated sites was associated with asthma. In high-SVI tracts, **eight hazards — predominantly $PM_{2.5}$ constituents and VOCs — were positively associated at FDR < 0.05**, and each one-quintile increase in the hazard mixture carried **35% higher asthma odds (95% CI 6–73%)**. Toluene, dry-cleaning proximity, PM organic matter and carbon monoxide each contributed over 10% of the joint association. Hazard correlations were themselves stronger in high-SVI areas, which is both the substantive finding and the analytical problem: with collinearity differing by stratum, the g-computation weights are not directly comparable across strata. Cross-sectional, self-reported physician-diagnosed asthma, residence-only exposure.

doi: 10.1016/j.jhazmat.2026.143083

Respiratory & allergic 2 records

Yue et al. 2026 Front Public Health

Short-term traffic-related gaseous pollutant exposure and childhood bronchopneumonia hospitalization in Guangzhou

20,623 paediatric bronchopneumonia admissions (ICD-10 J18, ages 0–14) at a single tertiary hospital in Huadu District, Guangzhou, 2014–2023, against daily NO_2 and SO_2 in a quasi-Poisson generalised additive model. Per 5 $\mu g/m^3$, peak single-day excess risk at lag 4 was **2.74% (95% CI 1.73–3.75) for NO_2** and **8.81% (4.94–12.82) for SO_2**; over the 0–7-day moving average these rose to **6.83% (4.93–8.77)** and **17.21% (10.18–24.70)**. Children aged 6–14 and boys were consistently more susceptible. Two problems: no particulate term is reported at all, so the gaseous estimates absorb whatever correlated PM signal exists and cannot be read as pollutant-specific; and a single-hospital catchment in a ten-million-person megacity leaves the denominator undefined and exposed to referral-pattern drift over a decade. Included because the lag structure and the age–sex susceptibility gradient are informative, not because the point estimates are attributable.

doi: 10.3389/fpubh.2026.1885590

Zhu et al. 2026 Medicine (Baltimore)

Air pollution and pneumonia hospitalization in Qingyang, China: a time-series analysis

Five years of pneumonia admissions (2015–2019) in an inland Chinese energy city against six pollutants, generalised additive plus distributed-lag nonlinear models. Per 10 $\mu g/m^3$: **$PM_{2.5}$ RR 1.004 (0.991–1.017)** and **PM_{10} 1.004 (0.999–1.010)** — both null — against **NO_2 1.221 (1.174–1.271)**, **SO_2 1.035 (1.022–1.048)**, and CO 1.382 (1.230–1.552) per 1 mg/m^3. Effects peaked at lag 1 for $PM_{2.5}$ and lag 0–7 for the gases; men were more vulnerable to SO_2, women to CO, and children $\leq$14 formed the high-risk group. The single-pollutant framing is the limitation — in a heating-dominated city $PM_{2.5}$, SO_2, NO_2 and CO share a common combustion source and a common seasonal cycle, and no multi-pollutant model is presented, so the apparent NO_2 dominance is a statement about correlation structure rather than about which pollutant acts. Useful mainly as a demonstration that PM mass can be the null term in its own time-series.

doi: 10.1097/MD.0000000000049962

Mechanistic toxicology 2 records

Balu et al. 2026 J Biochem Mol Toxicol

SRM 1650b administration to isolated rat heart aggravates ischemia-reperfusion injury via mitochondrial dysfunction

Langendorff-perfused male Wistar hearts exposed to NIST SRM 1650b heavy-duty diesel particulate at 10, 100 and 300 $\mu g/mL$ before 30 min ischemia and 60 min reperfusion. Dose-dependent deterioration of haemodynamics and elevated tissue injury, with increased oxidative stress in tissue and isolated mitochondria, reduced bioenergetic enzyme activity and respiratory efficiency, downregulated *Pgc-1α*, *Tfam*, *Polg*, *Fis1*, *Mfn1* and *Pink1*, lower mtDNA copy number, and suppressed PI3K/Akt signalling that fell further after the ischemic challenge. The design isolates a direct cardiomyocyte effect cleanly — no autonomic, pulmonary or systemic inflammatory route is available in an ex-vivo preparation — which is both its strength and its ceiling. The concentrations are far above any inhaled dose delivered to myocardium in vivo, and direct coronary perfusion bypasses the alveolar barrier entirely, so this establishes a mechanism that *can* operate, not one that does at ambient exposure.

doi: 10.1002/jbt.71038

Pan et al. 2026 Front Pharmacol

Effects of air pollution exposure on nasal mucosal immune-inflammatory markers in animal models of allergic rhinitis

PRISMA systematic review and random-effects meta-analysis of 18 rodent studies, standardised mean differences for nasal mucosal markers. Air pollutant exposure was associated with increased eosinophils and elevated IL-4, IL-5, IL-13 and OVA-specific IgE — a coherent enhancement of type 2 inflammation — with scattered evidence of epithelial barrier loss (reduced ZO-1) and innate activation (IL-1β, NLRP3). IFN-γ showed no consistent direction. Heterogeneity was substantial and the authors decline to claim a causal pathway. The pooling is the weak point: 18 studies

spanning diesel exhaust particulate, $PM_{2.5}$, PM_{10} and mixed urban aerosol at uncontrolled doses and durations, in an ovalbumin-sensitisation model whose relationship to human allergic rhinitis is itself contested, cannot yield a transferable effect size. No dose–response meta-regression is attempted, which is the analysis that would have justified the synthesis. Value here is as a marker map for the type 2 axis, not as an effect estimate.
doi: 10.3389/fphar.2026.1870023

Other clinical endpoints — 2 records

Zhou et al. 2026 BMJ Open

Environmental exposome study for prostate cancer in elderly men from Western China

580 histologically confirmed localised prostate cancers after radical prostatectomy, matched 1:2 to 1,160 cancer-free controls from the WeEndPd cohort, with urinary bisphenols and phthalate metabolites by LC-MS/MS and geocoded five-year SO_2 and $PM_{2.5}$. Highest-quartile five-year SO_2 (9.37–28.79 $\mu g/m^3$) gave **OR 1.65 (95% CI 1.08–2.72)** against the lowest; MECPP, MEHHP, MEHP, BPA and BPZ were all higher in cases; the WQS mixture of endocrine-disrupting chemicals gave **OR 1.26 (1.20–1.33)**. The $PM_{2.5}$ term is reported but is not the headline. Two structural problems: urinary EDC metabolites have half-lives of hours to days and were measured at or after diagnosis, so they cannot index the aetiologically relevant window — this is reverse-causation-prone by construction; and cases are surgical patients from one tertiary centre, which selects on operability and on health-seeking behaviour that also tracks residential air quality.
doi: 10.1136/bmjopen-2025-110957

Knapova et al. 2026 Appl Psychol Health Well Being

Daily relationship between air pollution and physical activity in an industrial region

693 adults in the industrial Ostrava region of the Czech Republic, 9,702 person-days, device-measured step counts against daily PM_{10} over 14 days, multilevel Bayesian models with within-person centring and adjustment for temperature, sunshine and precipitation. On days when PM_{10} exceeded a participant's own 14-day mean, step counts fell ($b = -0.027$, 95% CI -0.047 to -0.006); **a 1-SD PM_{10} increase corresponded to 361 fewer steps**. Self-reported tendency to monitor air quality did not moderate the effect, which argues the response is not deliberate avoidance behaviour. Exposure is ambient monitor data assigned to residence, not personal, so the estimate confounds pollution with unmeasured weather and visibility cues; 14 days is short for a within-person design; and non-probabilistic quota sampling limits transport. The finding matters for exposure modelling regardless: **behavioural response makes time-activity patterns endogenous to concentration**, which biases any exposure model that treats mobility as exogenous.
doi: 10.1111/aphw.70186

Burden, policy & mitigation — 1 record

Qian et al. 2026 J Hazard Mater

Stronger actions are needed to advance $PM_{2.5}$ mitigation and its health benefits under population aging in China

A new daily 1 km $PM_{2.5}$ product for mainland China 2015–2024, used to track concentration, WHO AQG exceedance and population-weighted exposure, with attributable cardiovascular mortality and a decomposition isolating the ageing offset. National $PM_{2.5}$ fell **35.57 → 20.62 $\mu g/m^3$** (population-weighted -37.35%), and the area above the WHO interim target of 35 $\mu g/m^3$ collapsed from 51.07% to 1.49% — yet **over 99% of the population still lives above the 5 $\mu g/m^3$ guideline**. Attributable CVD deaths fell from 2.16 (1.64–2.58) million in 2015 to 1.55 (1.14–1.91) million in 2022, then **rebounded to 1.76 (1.30–2.15) million in 2024**. The share of health benefit cancelled by population ageing rose from around 40% before 2020 to **56.54% in 2024**, exceeding 70% in some subregions. The burden estimates inherit exposure–response functions from external cohorts, so the uncertainty intervals understate structural uncertainty, and the 1 km product's own error is not propagated into the mortality intervals. The demographic-offset framing is the contribution and it generalises beyond China.
doi: 10.1016/j.jhazmat.2026.143101

Reproductive & developmental — 1 record

Sezer et al. 2026 Birth Defects Res

Preconception and early pregnancy exposure to ambient air pollution and risk of cleft lip and/or palate

Hospital-based case-control study, 127 nonsyndromic CLP cases against 271 controls, with $PM_{2.5}$, PM_{10}, O_3, NO_x, SO_2 and CO assigned for the three months before conception and the first trimester. Higher O_3, $PM_{2.5}$ and PM_{10} were each associated with elevated CLP risk in single-pollutant models; **O_3 was the dominant contributor in the WQS mixture** and the only exposure stable across windows and model specifications, while traffic-related pollutants did not survive multi-pollutant adjustment. The study is underpowered for a mixture analysis — 127 cases against a six-pollutant WQS index invites weight instability, and no bootstrap weight distribution is reported — and hospital-based control selection risks referral bias. Read as hypothesis-generating. The methodological point that carries: with correlated pollutants and a small case series, the pollutant that wins the mixture is often the one with the widest exposure contrast, not the one with the strongest biology.
doi: 10.1002/bdr2.70102

Occupational & indoor exposure — 1 record

Park et al. 2026 Sci Total Environ

Environmental evaluation of wood-based remodeling in aging buildings: indoor air quality and life-cycle carbon

Mock-up chamber characterisation of softwood finishes plus field monitoring in an occupied childcare facility and an unoccupied university classroom — before remodelling, 24 h after, and at four weeks — covering formaldehyde, anthropogenic and biogenic VOCs, particulate matter and total airborne bacteria, with a life-cycle global warming potential assessment. α-Pinene dominated the biogenic emission profile. In the occupied, ventilated building PM and bacterial counts fell or stabilised and most VOCs declined within four weeks; in the unoccupied, poorly ventilated classroom **α-pinene rose from 28.7 to 1713.6 $\mu g/m^3$ over four weeks** — a 60-fold accumulation. Net GWP in that building fell to 37.7 $kgCO_2eq$, 9–11 times below non-wood alternatives, on biogenic carbon storage. With $n = 2$ buildings differing simultaneously in occupancy, ventilation and use, those factors cannot be separated, and the PM measurements are reported only qualitatively. The transferable point is the ventilation dependence: a material acceptable in one air-exchange regime is not in another, which argues for continuous indoor monitoring rather than commissioning-time spot checks.

doi: 10.1016/j.scitotenv.2026.182065

Corpus register

First author	Focus	Journal	Design	Metric	Region
Reproductive & developmental					
Sezer et al.	Preconception and early pregnancy exposure to ambient air pollution and risk of cleft lip and/or palate	*Birth Defects Res*	Case-control	$PM_{2.5}$ + PM_{10} + O3	Turkiye
Respiratory & allergic					
Yue et al.	Short-term traffic-related gaseous pollutant exposure and childhood bronchopneumonia hospitalization in Guangzhou	*Front Public Health*	Time-series	Traffic gases (NO2, SO2)	China
Zhu et al.	Air pollution and pneumonia hospitalization in Qingyang, China: a time-series analysis	*Medicine (Baltimore)*	Time-series	$PM_{2.5}$ + PM_{10} + 4 gases	China
Mechanistic toxicology					
Balu et al.	SRM 1650b administration to isolated rat heart aggravates ischemia-reperfusion injury via mitochondrial dysfunction	*J Biochem Mol Toxicol*	Ex vivo perfused organ	Diesel PM (SRM 1650b)	India
Pan et al.	Effects of air pollution exposure on nasal mucosal immune-inflammatory markers in animal models of allergic rhinitis	*Front Pharmacol*	Umbrella review / meta-analysis	$PM_{2.5}$ / PM_{10} / DEP (animal)	Global
Occupational & indoor					
Park et al.	Environmental evaluation of wood-based remodeling in aging buildings: indoor air quality and life-cycle carbon	*Sci Total Environ*	Field measurement	PM (indoor, unspeciated)	South Korea
Exposure assessment & modelling					
Chen et al.	Geospatial-based neighborhood hazards and asthma in adults: interaction with social vulnerability	*J Hazard Mater*	Exposome-wide association	$PM_{2.5}$ constituents + VOCs	USA
Nyoni et al.	Performance of road-traffic-based exposure proxies against personal $PM_{2.5}$ measurements in three Sub-Saharan African countries	*Air Qual Atmos Health*	Proxy validation vs personal exposure	$PM_{2.5}$ (personal, filtered)	The Gambia / Kenya / Mozambique
Burden, policy & mitigation					
Qian et al.	Stronger actions are needed to advance $PM_{2.5}$ mitigation and its health benefits under population aging in China	*J Hazard Mater*	Machine-learning model	$PM_{2.5}$	China
Other clinical endpoints					
Knapova et al.	Daily relationship between air pollution and physical activity in an industrial region	*Appl Psychol Health Well Being*	Panel study	PM_{10}	Czech Republic
Zhou et al.	Environmental exposome study for prostate cancer in elderly men from Western China	*BMJ Open*	Case-control	$PM_{2.5}$ + SO2	China

Provenance and method

Entry window **31 July 2026**, contiguous with the 30 July issue — no gap. Sources: **PubMed** (E-utilities and connector) queried on the `[Date - Entry]` field along two independent axes, health and instrumentation; **Europe PMC** `CREATION_DATE:[2026-07-31 TO 2026-07-31]`; **arXiv** relevance query with a local recency screen (none in window); **OpenAlex** returned HTTP 429 and remains unusable for date-windowed harvesting; **ClinicalTrials.gov** returned one record updated in the window — NCT05160948, personal-monitor result feedback as a behavioural intervention, observational, so no endpoint analysis was run — accumulated to `state/trials.json` for the weekly rollup. **11 records in scope.**

Scheduling correction. This issue was originally built with a two-day window (31 July–1 August) and published as the 1 August issue. That was wrong: papers entering PubMed on 1 August belong to the 1 August run. The window was re-cut to 31 July on the PubMed `entrez` entry date of every record. **Four records with entry date 2026-08-01** — Zhang ($PM_{2.5}$ components and cognition), Daoud (suicide mortality), Bello (prenatal $PM_{2.5}$, Quebec) and Pai (incense burning and SGA) — were *removed* from this issue, dropped from `state/seen.json` so they will be re-found, and stashed in `state/deferred_2026-08-01.json` for the 1 August run. Every summary retained here is unchanged from its original text.

Two records reached this issue through **Europe PMC** rather than PubMed and carry earlier PubMed entry dates (Knapova, 8 July; Balu, 21 July). They are dated to this issue because 31 July is when the Europe PMC creation-date window first surfaced them; a PubMed-only pipeline would never have recovered them. This is stated rather than hidden because it means the issue's window is not uniform across sources.

Known defect: zero sensing records for the second consecutive window. PubMed does not index *Atmos. Meas. Tech.* or *Atmos. Chem. Phys.*; the Consensus connector has no entry-date filter; OpenAlex date filtering is paywalled. The July monthly rollup quantified the cost — 23 sensing records for the month against 11 across all five daily issues. The fix remains a Crossref by-journal feed keyed on `created` for AMT (1867-8548) and ACP (1680-7324).

Subtopic, design, particle-metric and geography labels were assigned from title, abstract, keywords and MeSH terms; assignment is single-label by dominant contribution and is therefore a judgement, not a controlled vocabulary. Effect estimates in the forest plot are transcribed verbatim from abstracts and use heterogeneous exposure contrasts — see the figure caption. Every DOI was verified against the PMID's own PubMed record and resolved before publication. Content derived from PubMed and Europe PMC metadata; abstracts remain the copyright of their respective publishers.